

Test 18: Chiral symmetry AIII at $\nu=1/2$

Verdict: PASS | Generated: 2026-09-15 01:01:13 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT: $\|SHS+H\|/\|H\| \lesssim 1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

METHOD: EXACT: $\|SHS+H\|/\|H\| \lesssim 1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

Test 18 HARDCORE: exact 0.00e+00, ν -row 5.77e-03 (ν_{rms} 0.0058, in window), vortex 0.00e+00 (preserved), $S^2=I$ 0.00e+00 (96x96 torus).

Result [HARDCORE pass 2]: 4 sub-checks, 0 failed $\Rightarrow$ PASS

- 'plots/plot_01.png' $\Rightarrow$ 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' $\Rightarrow$ animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 18 --no-two-pass`

Independent verification guide

See computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[01:00:19.945] [+4410.973s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, $t=1.00$, $W=0.00$, torus, 0 vortices, model=:monumental

[01:00:35.975] [+4427.003s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, $t=1.00$, $W=1.00$, torus, 0 vortices, model=:monumental

FULL COMPUTATION LOG

[01:00:56.095] [+4447.123s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, $t=1.00$, $W=0.00$, torus, 2 vortices, model=:monumental

??

TEST 18 (pass 2, HARDCORE) - chiral symmetry AIII $\Rightarrow$ deep audit

CONSOLE CAPTURE

Exact row + ν_{rms} quantitative contract + vortex-preservation row +

$S^2 = I$ operator sanity (launch-standard lattice)

??

ν RUNTIME: 3 builds + 3 S^2+H^2 defect evals at 96x96 (Diagonal S , $O(N^2)$) $\Rightarrow$? ? 1?2 min

H1 exact row ($\nu=1/2$, $W=0$): defect = 0.00e+00 < ? 1e-10

H2 ν row ($W=1.0$, $Nv=0$): defect = 5.77e-03 vs ν_{rms} = 0.0058 (analytic) ? [0.5, 2.0]x?

H2v vortex row ($Nv=2$, $W=0$): defect = 0.00e+00 < 1e-10 ? (bond phases PRESERVE AIII)

H3 operator sanity: S^2 ? I^2/I ? = 0.00e+00 < ? 1e-12

??

exact row < 1e-10 PASS ? $SHS+H^2/H^2$ = 0.00e+00 at $\nu=1/2$, $W=0$

ν_{rms} contract ($W=1.0$) PASS ? defect 5.77e-03 vs analytic 2???_F/?H?_F = 0.0058 (size-independent)

vortices PRESERVE AIII PASS ? defect = 0.00e+00 ($Nv=2$, $q=-1.00$, $W=0$) ? bond phases; Byers-Yang for integer q

S? = I PASS ? ?S??I?/?I? = 0.00e+00

Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS